Q.22 Write and explain the algorithm for Binary Search. (CO3)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain Traversal techniques of Binary Trees. Discuss Inorder Traversal with example. (CO3)

Q.24 Describe Heap Sort in detail with algorithm, example, and complexity. (CO3)

Q.25 Explain in detail: Divide and conquer, Greedy, Dynamic Programming, Backtracking. (CO4)

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Roll No. ....

4th Sem / Artificial Intelligence & Machine Learning

Subject : Algorithm Design Techniques

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which of the following is not a property of an algorithm? (CO1)

a) Finiteness b) Definiteness

c) Versatility d) Output

Q.2 The time complexity of inserting an element at the end of an array is______. (CO2)

a) O(1) b) O(log n)

c) O(n) d) O(n log n)

Q.3 Which data structure is used in Depth First Search? (CO3)

a) Stack b) Queue

c) Heap d) Linked List

Q.4 Which sorting algorithm is *in-place* and *unstable*?

a) Merge Sort b) Heap Sort (CO3)

c) Counting Sort d) Radix Sort

Q.5 Bellman-ford algorithm can handle_____ . (CO4)

a) Only positive weights

b) Only 0 weights

c) Negative weights

d) Floating weights only

Q.6 Dynamic Programming is mainly used for solving problems with_____. (CO4)

a) Independent Subproblems

b) Overlapping subproblems

c) Unrelated data

d) Backtracking

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Define an Array. (CO1)

Q.8 Name the two operations performed on a stack. (CO1)

Q.9 Define average case complexity. (CO2)

Q.10 Write the recurrence relation for Merge Sort. (CO2)

Q.11 What is Greedy Algorithm? (CO3)

Q.12 Write any two applications of Graphs. (CO3)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 What is an algorithm? Explain its characteristics. (CO1)

Q.14 Explain worst-case and best-case complexities with examples. (CO2)

Q.15 Write a short note on “Space Complexity”. (CO2)

Q.16 Compare Array and Linked List. (CO2)

Q.17 Explain Greedy Method with a suitable example. (CO3)

Q.18 Write the algorithm for deleting a node from a Binary Search Tree. (CO3)

Q.19 Explain the concept of memoization in Dynamic Programming. (CO4)

Q.20 Describe Prim’s algorithm. (CO4)

Q.21 Explain Selection Sort with an example. (CO3)